Lateral analysis - Level 1 - First rotor critical speed (simple calculations using standard equation)

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1 Purpose.

This document describes how to perform calculation of the first (lateral) critical speed of a pump rotor.

The procedure to define what level of detail for the dynamic analysis is required to be in accordance with good practice of pump design is described in G.10.00.09 and *ANSI/HI 9.6.8-2014*: if calculated RUN is higher than 15 and less equal to 20, a **Lateral Analysis - Level 1 – First rotor critical speed (simple calculation using standard equation)** is required.

A calculation sheet called Mod.423 and a python program critical speed.py are available to evaluate both the static deflection and the first critical speed of a shaft with multiple masses. It is created for shaft with multiple masses attached, but it works properly also for single mass.

2 Abbreviations.

AF	amplification factor[1]
BB	between bearing (API610)
DOF	degree of freedom
OH	overhung (API610)
R	risk number (ANSI HI 9.6.8-2014)
RUN	R multiplied by U (ANSI HI 9.6.8-2014)
U	uncertainty value (ANSI HI 9.6.8-2014)

3 Scope.

This procedure is mainly applicable to OH and BB pumps.

[1] The Amplification Factor is the ratio between the dynamic and the static dynamic response to a harmonic excitation. It is related to the system damping: higher damping means lower AF and vice versa.

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4 General procedure: description.

Level 1 calculations involve significant simplifying assumptions (e.g., infinite bearing stiffness, dry running rotors, no gyroscopic effect, …). Depending on the type of analysis being performed, these assumptions can introduce significant errors. Although useful for initial sizing and refinement of the design, such methods are usually too coarse for the results to be used to qualify a final design against a particular standard or program.

Depending on the design, installation and operating speed of the pump these assumptions may render the calculation completely invalid. It is therefore necessary for the designer to confirm the expected level of error attendant when simple calculations are used.

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5 Methodology, analysis and interpretation of results.

Methodology.

For these calculations simple equations are derived from vibration theory: mass and spring system subjected to the condition of undamped free vibration for single degree of freedom are studied.
In a simple system having linear spring and translational motion, a mass *m* suspended from a spring of stiffness *k,* having a static linear deflection *δ* under its own weight *W* has a natural frequency that is function of *δ.*

$$N_c = \frac{60}{2\pi}\sqrt{\frac{k}{m}} = \frac{60}{2\pi}\sqrt{\frac{W}{m\delta}} = \frac{60}{2\pi}\sqrt{\frac{g}{\delta}}$$

$$N_c = critical\ speed\ [cpm]$$

$$k = stiffness\ [N/m]$$

$$m = mass\ [kg]$$

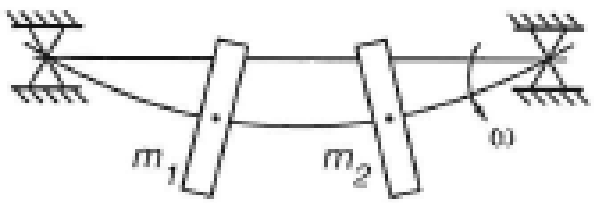
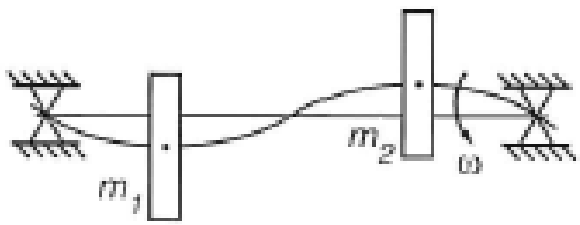
$$W = weight\ [N]$$

$$\delta = static\ deflection\ [m]$$

$$g = 9.8066\ [{}^{m}/_{s^2}]$$

It is important to note that any object made of elastic material has infinite natural frequencies of vibration that can be excited by residual imbalance at a frequency synchronous with running speed, because the pump rotating assembly is not uniform around the centerline of the shaft. There are variations in the density of the materials as well as manufacturing tolerances and casting irregularities contributing to this issue. The lowest natural frequency excitable by imbalance such that its amplification factor is greater than 3.33 is known as the rotor first critical speed.

At the first critical speed the shaft will bend to the simplest shape possible; at the second critical speed it will bend to the second simplest shape possible, etc…

First critical speed	Second critical speed
	

For level 1, only the first critical speed will be considered. Although ignored, gyroscopic effects in the rotor may increase or decrease the critical speed somewhat versus the non-rotating speed.

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1) Rotor with a single attached mass.

If the shaft mass is small compared to the attached mass, the first critical speed is obtained by the following relationship:

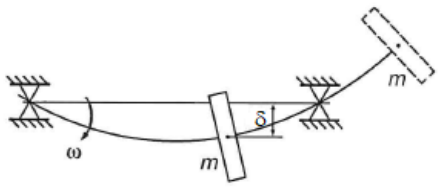
$$N_c = \frac{60}{2\pi}\sqrt{\frac{g}{\delta}}$$

$$N_c = critical\ speed\ [cpm]$$

$$\delta = static\ deflection\ [m]$$

$$g = 9.8066\ [{}^{m}/_{s^2}]$$

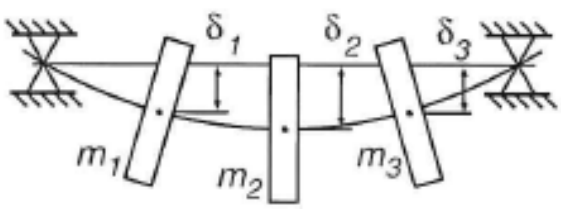
The relationship is independent of shaft inclination (horizontal, vertical or intermediate); δ represents shaft deflection, during rotation, at the mass location. The mass may be between bearings or overhung (dashed in the picture).

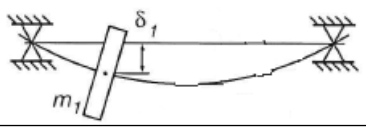


2) Shaft of negligible mass carrying several concentrated masses.

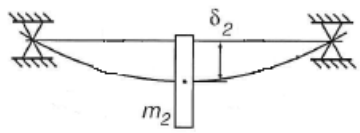
Rayleigh-Ritz equation and Dunkerly equation provide a value for the first critical speed.

Rayleigh-Ritz	Dunkerly
$\omega_c = \sqrt{\dfrac{g\sum_1^j M_n\delta_n}{\sum_1^j M_n\delta_n^2}}$ $\omega_c = critical\ speed\ [rad/s]$ $M_n = static\ weight\ of\ n^{th} mass\ [kg]$ $\delta_n = static\ deflecetion\ at\ n^{th} mass\ [m]$ $j = total\ number\ of\ masses\ [-]$ $g = 9.8066\ [{}^{m}/_{s^2}]$	$\dfrac{1}{\omega_c^2} = \dfrac{1}{\omega_1^2} + \dfrac{1}{\omega_2^2} + \dfrac{1}{\omega_3^2} + \ldots$ $\omega_c = critical\ speed\ for\ the\ rotor\ [rad/s]$ $\omega_1 = critical\ speed\ first\ mass\ only\ [rad/s]$ $\omega_2 = critical\ speed\ second\ mass\ only\ [rad/s]$ $\omega_3 = critical\ speed\ third\ mass\ only\ [rad/s]$





$$\omega_1 = \sqrt{\frac{g}{\delta_1}}$$



$$\omega_2 = \sqrt{\frac{g}{\delta_2}}$$

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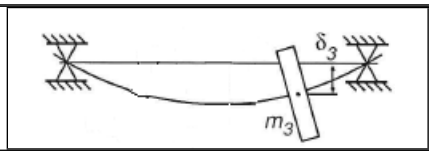
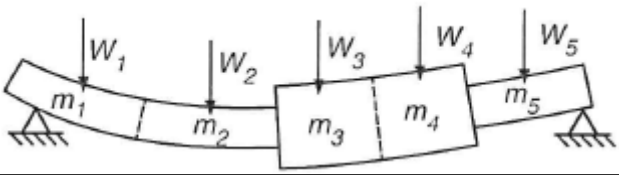
		$\omega_3 = \sqrt{\frac{g}{\delta_3}}$
Overestimates natural frequencies	Underestimates natural frequencies	

3) <u>Shaft without attached mass.</u>

Rayleigh-Ritz equation can be used to determine the first natural frequency: subdividing the shaft into several pieces with its own mass, it is possible to treat each part as a concentrated mass at its centre of gravity. The number of subdivisions to use is a matter to learn with experience, but it will be found that good level of accuracy is obtained with rather crude approximations.



4) <u>Shaft with its own mass carrying several concentrated masses (corrected Dunkerly equation)</u>

Previous method can be used to evaluate shaft (alone) critical speed: this value can be included in Dunkerly equations to correct the natural frequency evaluation.

$$\frac{1}{\omega_c^2} = \frac{1}{\omega_{shaft}^2} + \frac{1}{\omega_1^2} + \frac{1}{\omega_2^2} + \frac{1}{\omega_3^2} + \dots$$

6 Acceptance Criteria.

In case the first dry natural frequency (first dry “critical speed”) is at least 30% above the maximum continuous operating speed (40% in case rotor is designed to run dry), then the rotor is considered to be classically rigid (ref. API 610).

Classically rigid horizontal rotors are expected to have design integrity with regards to lateral rotordynamic behavior.

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References

Centrifugal and Axial Flow Pumps – Theory, Desigh and Application – A.J. Stepanoff

ANSI/HI 9.6.8-2014 Rotordynamic Pumps – Guideline for Dynamic Pumping Machinery

Shigley's Mechanical Engineering Design – R.G. Budynas J.K. Nisbet

G.10.00.09 - Guidelines for dynamic analysis complexity level definition

Mod.423_Rev0_Shaft critical speed calculation

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Annex A: Lateral analysis - Level 1 - First lateral critical speed calculator module

A calculation sheet called Mod.423 is available to simplify the calculation. It is necessary to create a local working directory with read/write permissions and to copy the calculation sheet and the python module inside it.

STEP 1

Define material properties.

shaft material	
E[GPa]	199.948
ρ [kg/m³]	7833.41

STEP 2

Create the shaft, subdividing it in sections with variable diameter, following the rule length over diameter ratio lower than one. A node (connection between the shaft section) must be defined:

1. where concentrated masses are attached
2. where bearings are placed
3. where it is important to know the static deflection.

Maximum elements number is 80.

Shaft		
element ID	L [mm]	d [mm]
1	100	100
2	100	100
3	100	100
4	100	100
5	100	100
6	200	200
7	200	200
8	200	200
9	200	200
10	200	200
...	...	...

STEP 3

Indicate bearings stations/nodes.

	node ID	s [mm]	δ [mm]
first bearing	3	200	0
second bearing	34	5800	0

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STEP 4	Indicate where concentrated masses (impeller, coupling, etc…) are placed.

concentrated mass			
Node ID [-]	M [kg]	s [mm]	δ [mm]
16	500	2500	0
21	500	3500	0

STEP 5	Press the button “create input file”

create input file

STEP 6	Create graphical representation of the shaft with the button “graph”

graph

STEP 7	Go to "Start" in windows and digit "Anaconda Prompt"
STEP 8	In anaconda prompt window, go to the working directory using dos commands (cd, cd.., etc...)
STEP 9	In anaconda prompt window, digit python “Mod.423 – Critical Speed Calculator.py" and press enter. In the working directory two result files will be created: • static_deflection.txt; • critical_speed.txt.
STEP 10	Create deflection diagram and import results (press button "read results)

read results

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STEP 11	These worksheets calculate the first lateral critical speed of a shaft following four different methods: • Rayleigh's equation (massless shaft); • Dunkerly's equation (massless shaft); • Rayleigh's equation (shaft isolated); • Dunkerly's equation "corrected" (include the mass of the shaft). The four results are summarised in the sheet

Rayleigh's equation	
First critical speed (massless shaft) [rpm]	779.5336
Dunkerly's equation	
First critical speed (massless shaft) [rpm]	768.2722
Rayleigh's equation	
First critical speed of the shaft [rpm]	744.3472
Dunkerly's equation	
First critical speed [rpm]	534.5911

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